4.1.1 Set Operations:

Algorithm:

Step 2: Input elements of Set A

Step 3: Input elements of Set B

Step 4: Display Union

set_a | set_b

Step 5: Display Intersection

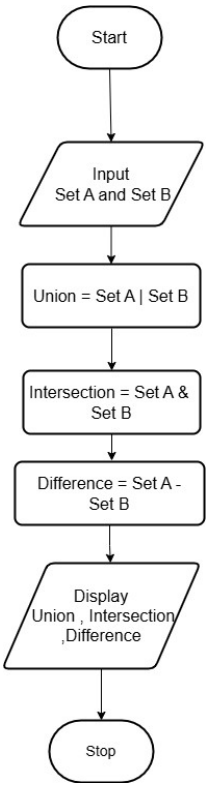
set_a C set_b

Step 6: Display Difference

set_a - set_b

Step 7: Stop

Flowchart:



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4.1.1. Set Operations

02:55

Write a Python program to perform union, intersection and difference operations on $Set\ A$ and $Set\ B$.

Input Format:

- First Line prompts "Set A: " followed by space-separated list of integers for $Set\ A$.
- The second input prompts "Set B: " followed by space-separated list of integers for $Set\ B$.

Output Format:

- The first line prints "Union: " followed by the union of $Set\ A$ and $Set\ B$.
- The second line prints "Intersection: " followed by the intersection of $Set\ A$ and $Set\ B$.
- The third line prints "Difference: " followed by the difference of $Set\ A$ and $Set\ B$.

Note:

- If there is no intersection between the two sets, the program prints an empty set, which appears as "set()" in the output.
- Please refer to the visible test cases for better understanding.

Sample Test Cases +

Explorer

setoperat...

Submit

Debugger

```python

a = set(map(int, input("Set A: ").split()))

b = set(map(int, input("Set B: ").split()))

print("Union:", a | b)
print("Intersection:", a & b)
print("Difference:", a - b)


```

Terminal Test cases